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Project: Generative AI-Based Ligand Binding Pose Refinement from Physics-Based Priors

Report 1: Understanding of Proteins, Substrate and Molecular Docking

Topics:

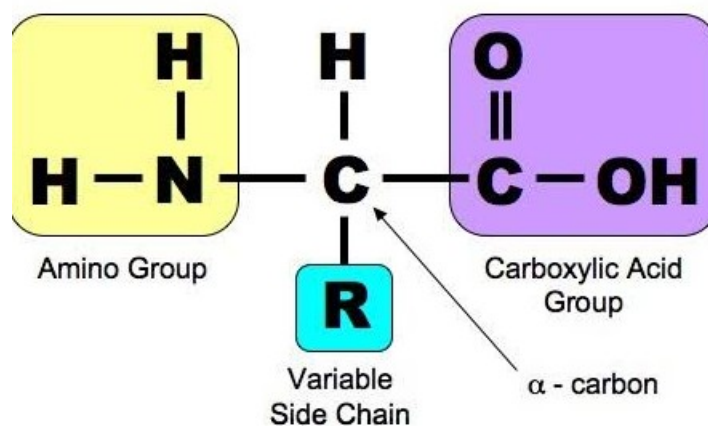
- 1) Proteins (Amino Acids)
- 2) Substrate
- 3) Molecular Docking

I PROTEINS:

- (i) Proteins transport nutrients, catalyze chemical reactions and are the structural components of living organisms.
- (ii) Proteins are made up of 20 set of standard amino acids.

AMINO ACIDS:

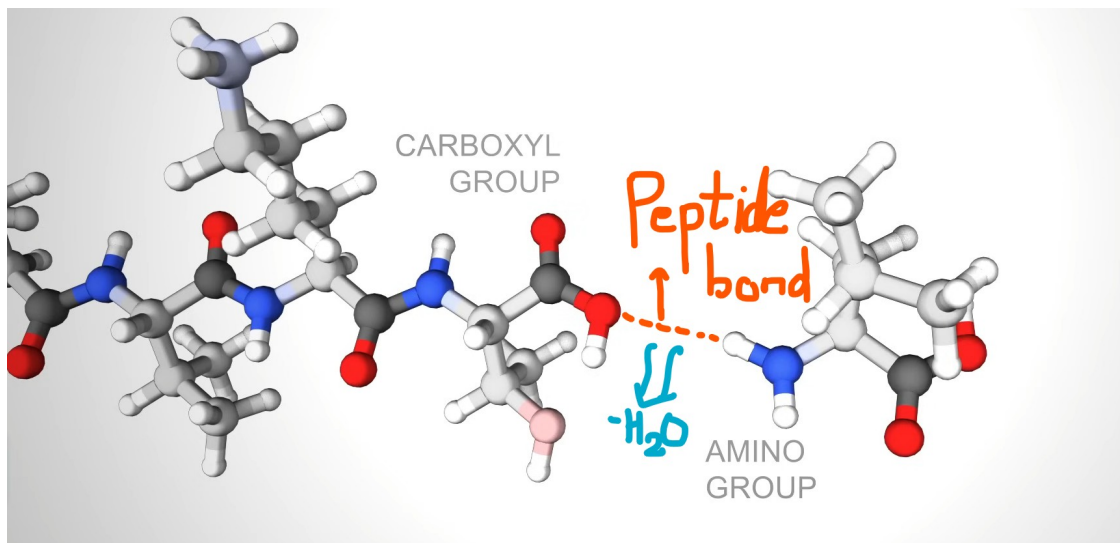
- (i) They are mainly composed of carbon, oxygen, nitrogen and hydrogen.
- (ii) Each amino acid has three key parts attached to a central (alpha) carbon atom:
 - a. An amino group (-NH₂)
 - b. A carboxyl group (-COOH)
 - c. A side chain (R group) – the only part that varies between amino acids and determines their chemical properties.

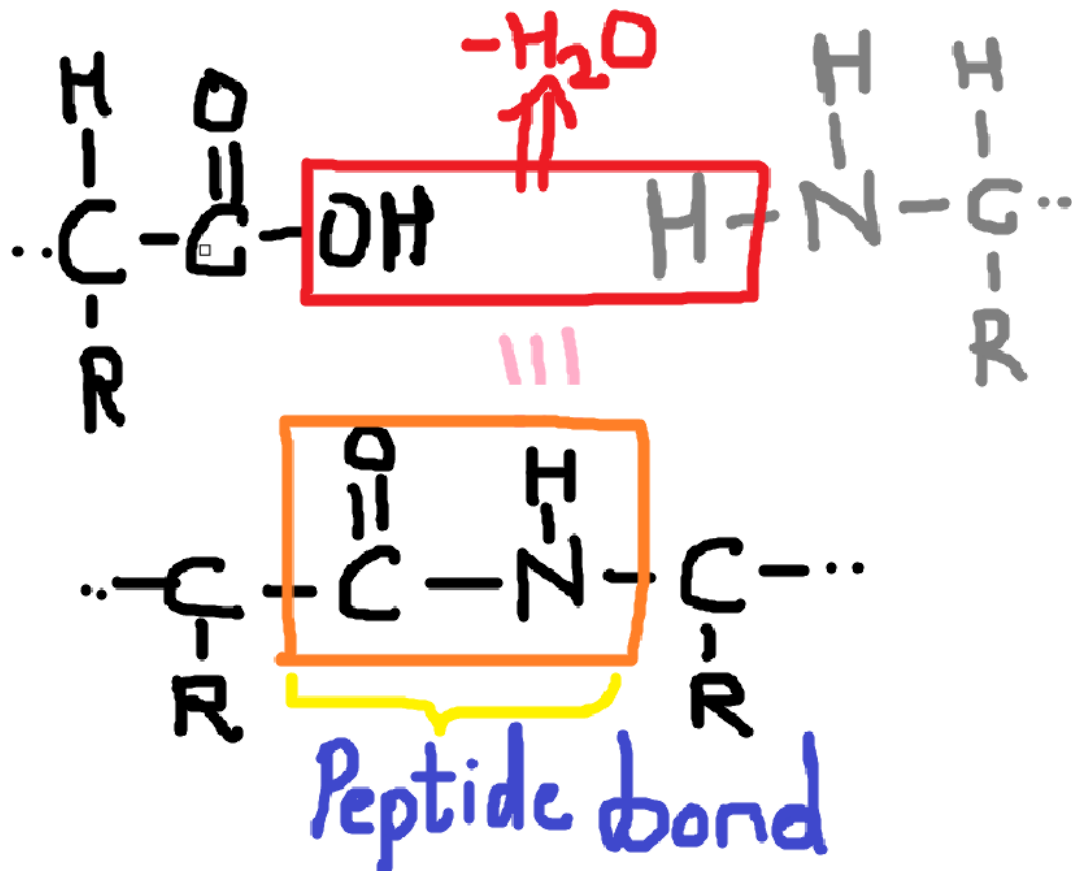


- (iii) Amino Acids are classified by their side chain (-R group) structure.
 - a. Hydrophobic: The side chain that repel water molecules.
 - b. Hydrophilic: The side chain which favorably interact with water molecules (through hydrogen bonding).
 - c. Charged: Interact with oppositely charged molecules/amino acids.

PRIMARY STRUCTURE OF PROTEINS:

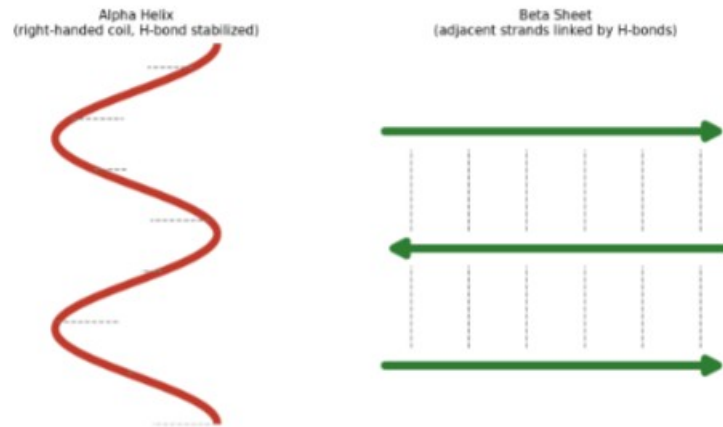
- (i) The primary structure of proteins is formed due to the linear sequential joining of amino acids as encoded by the DNA.
- (ii) The amino acids are joined via peptide bonds.





SECONDARY STRUCTURE OF PROTEINS:

- (i) Alpha helix: It's a right-handed coil stabilized by hydrogen bonds between nearby amine and carboxylic groups.
- (ii) Beta sheet: It is formed when hydrogen bonds stabilize two or more adjacent strands.



TERTIARY STRUCTURE OF PROTEINS:

The tertiary structure is the overall 3D shape of a single protein chain, determined by the properties of its amino acids.

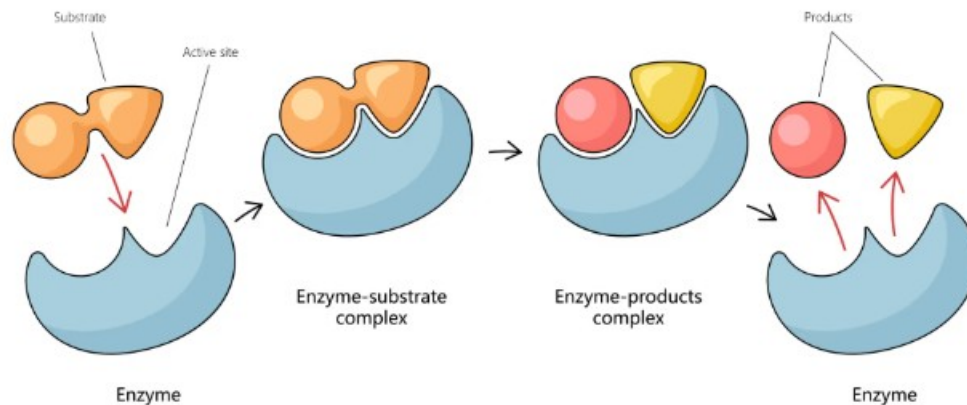
- (i) Many proteins fold into globular shapes with hydrophobic side chains tucked inside, shielded from surrounding water.
- (ii) Membrane-bound proteins instead cluster hydrophobic residues on the outside, allowing them to interact with lipid membranes.
- (iii) Charged amino acids let proteins interact with molecules bearing complementary (opposite) charges.

II PROTEIN-LIGAND INTERACTION

ENZYME-SUBSTRATE INTERACTIONS:

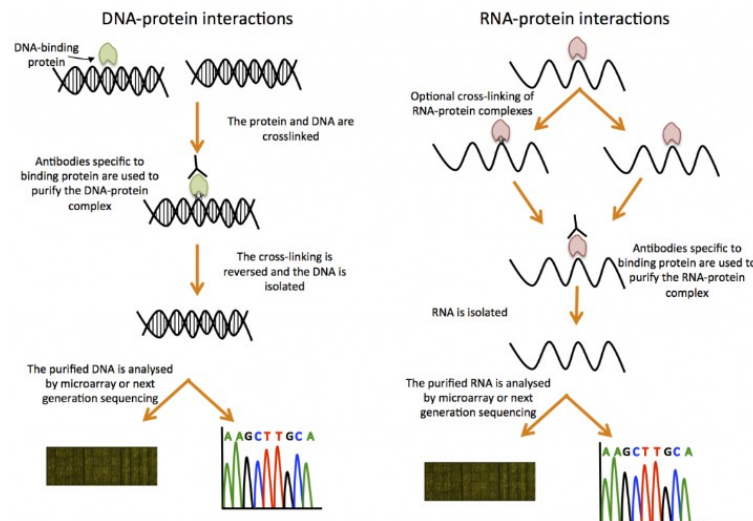
- i. Enzymes are special proteins with a specific part typically referred to as the *"lock"* (active site).
- ii. Substrates are the *"keys"* (molecules) that fit into the *lock*.
- iii. When a substrate binds to an enzyme, it undergoes a chemical reaction and is converted into a product.
- iv. Enzyme releases the product after the reaction and returns to its original shape.

Mechanism of Enzyme Action



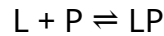
PROTEIN-NUCLEIC ACID INTERACTIONS:

- Proteins can connect to nucleic acids (DNA and RNA) to regulate gene expression, DNA replication, and RNA processing.
- Some proteins can bind to specific spots in the DNA and prevent those spots from being “read”, ensuring that genes are activated or suppressed when needed.



BINDING AFFINITY AND DISSOCIATION CONSTANT:

- i. Binding affinity is the measure of how tightly a ligand binds with the active site.
- ii. Suppose we represent ligand as {L} and protein as {P} then when they are bound together we will represent them as {LP}. This is a two-state process and can be represented as:



- iii. The dissociation constant can be calculated as:

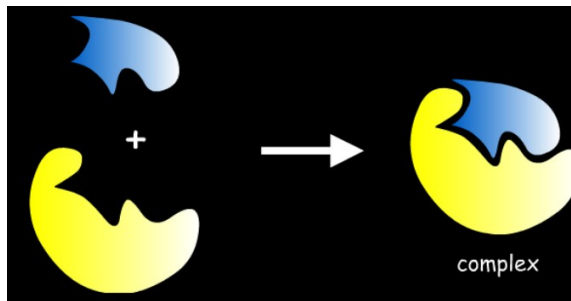
$$k_D = \frac{[L][P]}{[LP]}$$

- iv. A lower dissociation constant means higher binding affinity and a higher dissociation constant means lower binding affinity.
- v. Ideal drug candidates have higher binding affinity to intended targets.

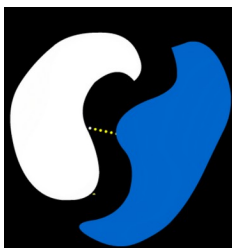
III MOLECULAR DOCKING

UNDERSTANDING DOCKING

- i. Molecular recognition is the ability of biomolecules to recognize other biomolecules and selectively interact with them in order to promote fundamental biological events this is known as Molecular Docking.
- ii. Types of Molecular Docking:
 - a. Lock and Key Theory: A substrate fits into the active site of a macromolecule, just like a key fits into a lock.



- b. Induced-Fit Theory: Both ligand and target mutually adapt their structure until an optimal fit is achieved.

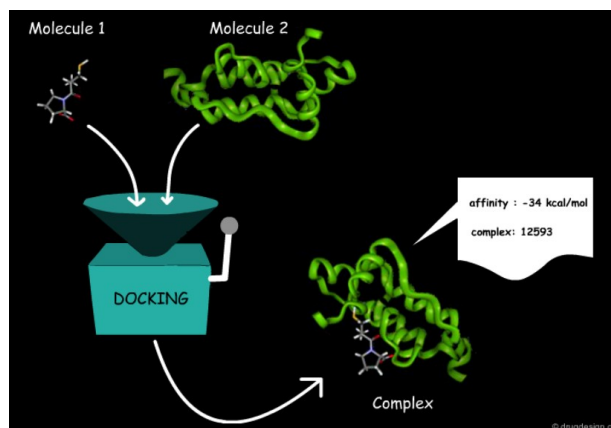


COMPUTATIONAL DOCKING:

- i. It is a computational science aiming at predicting the optimal binding pose of interacting molecules.
- ii. Most common docking systems: protein-protein, nucleic acid-protein and protein-small molecule.

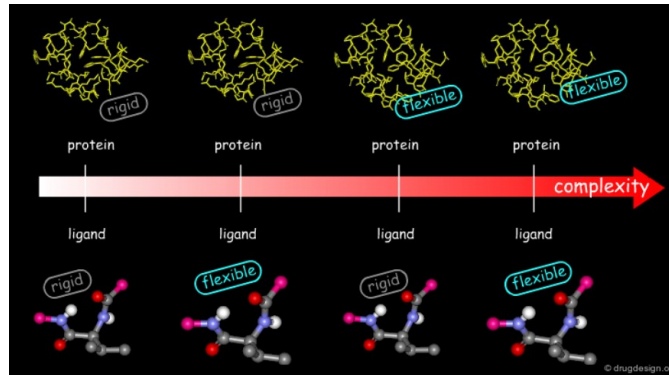
DOCKING PROBLEM:

- i. The task of predicting how molecules interact with each other is called docking problem.
- ii. Molecular docking software is like a black box whose input is set of 3D structures to be docked and output is the predicted complexes of molecules.
- iii. The main goal is to find the 3D structure of the complex and binding affinity.



CLASSIFICATION OF DOCKING PROBLEMS

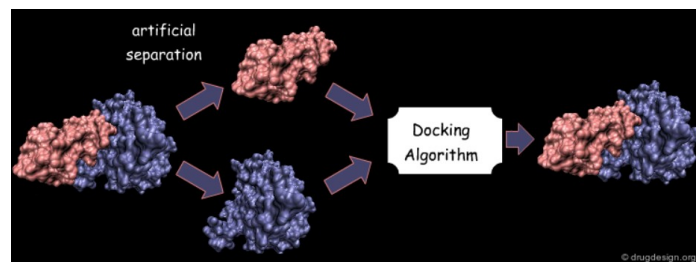
By Molecular Flexibility:



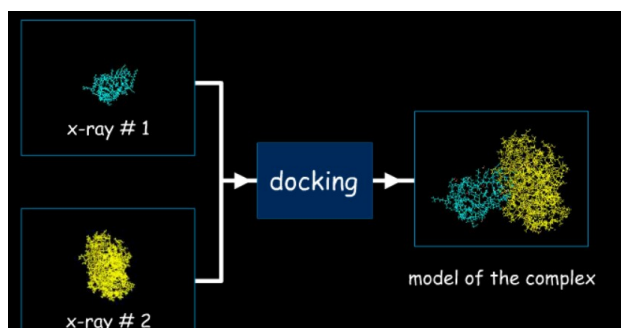
By Availability of Structural Data:

Docking difficulty also depends on what structural information is available in the start of the process.

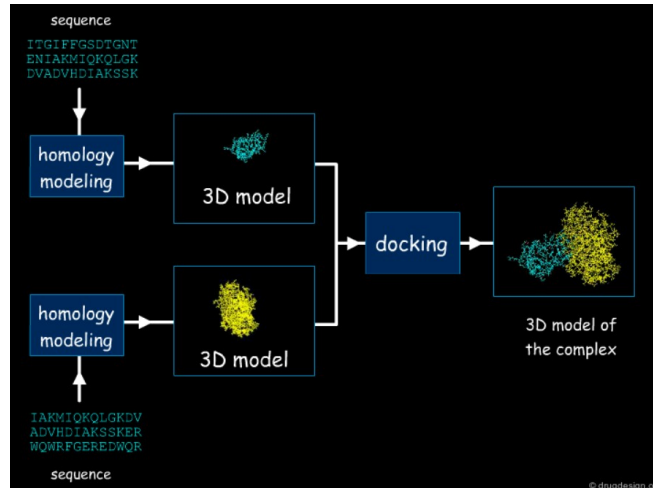
- i. Bound Docking: The experimental structure of complex is already known. Used mainly to calibrate the software.



- ii. Unbound Docking: Only the separate molecule structures are known. You'll have to correctly develop the model for this condition.



- iii. Modeled Docking: No experimental structure at all. It relies on homology modeling.



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